

Task: Year 10 Term 1 Week 4 test prep

1. Expand and simplify:

$$4(y + 3) - 5y + 6$$

2. Expand and simplify:

$$c(c + 7) + 5(c + 6)$$

3. Expand and simplify the following:

$$(m - 2)(m + 9)$$

4. Expand the following using binomial expansion:

$$(x + 3)(x + 10)$$

5. Factorise the following expression by taking out the highest common factor:

$$14pqr - 5pst$$

6. Factorise the following expression by taking out the highest common factor:

$$31x - x^2$$

7. Express $\frac{4x}{4}$ in simplest form.

8. Simplify the expression $\frac{2t^3}{3t^3}$.

9. Rewrite $\frac{-3}{k}$ with $5k$ as the denominator.

$$\frac{-3}{k} = \frac{\boxed{}}{5k}$$

10. Simplify $\frac{3x}{6} + \frac{4x}{6}$

11. Simplify $\frac{3x}{35} - \frac{2x}{15}$.

12. Simplify $\frac{c}{4} \times \frac{d}{3}$.

13. Simplify the following:

$$\frac{10}{9y} \times \frac{11}{9y}$$

14. Simplify $\frac{-7x}{8} \div \frac{9x}{10}$.

15. The product of two fractions is 1. If one of the fractions is $\frac{5x}{9y}$, what is the other fraction?

Let F be the other fraction.

16. Simplify $p^5 \div p^3 \times p^4$.

17. Simplify $\frac{y^2 \times y^5}{y^3 \times y^7}$
